VIREN VASANTPURI VAIRAGI

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LinkedIn | GitHub | Portfolio[https://viren501.github.io/my-portfolio/]

EDUCATION

S.P.B Patel Engineering College — Mehsana, Gujarat
Bachelor of Engineering in Computer Engineering (CGPA: 8.00) — *2022 – 2026*

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Software Engineering, Database Management Systems.

EXPERIENCE

Brainy Beam Info-Tech Pvt. Ltd. — Ahmedabad, Gujarat
Data Science & Machine Learning Intern — *01/2026 – 03/2026*

- Accomplished a robust predictive job offer application as measured by successful model evaluations during the internship lifecycle, by managing the complete Data Science pipeline.
- Engineered precise predictive behavior by mapping target variable factors where 'Joined' equals 1 and 'Not Joined' equals 0 during preprocessing.
- Architected and deployed the definitive model to a live Streamlit Cloud application, enabling real-time recruitment optimization.
- Benchmarked multiple classification algorithms, conclusively establishing Random Forest as the superior performer over alternatives like Gradient Boosting.

InfoLabz IT Services Pvt. Ltd. — Ahmedabad, Gujarat
Data Analysis & Machine Learning Intern — *07/2025 – 07/2025*

- Accomplished clean data-driven pipelines as measured by improved model evaluation readiness, by applying data preprocessing and EDA techniques.
- Authored supervised learning evaluations utilizing Python, Pandas, and Scikit-Learn libraries.

PROJECTS

Job Offer Acceptance Prediction — *Python, Scikit-Learn, Random Forest, Streamlit*

- Accomplished advanced recruitment forecasting as measured by optimized HR strategies and reduced candidate dropouts, by developing an end-to-end Machine Learning pipeline.
- Engineered data features and behavioral indicators by applying Data Cleaning and EDA on salary expectations.
- Deployed a fully functional web application using Streamlit to provide real-time forecasting, leveraging the optimal Random Forest algorithm.

Adaptive & Collaborative Traffic Control System — *AI/IoT, Random Forest, V2X*

- Accomplished real-time traffic density monitoring as measured by successful congestion forecasting, by integrating IoT sensor data with Machine Learning algorithms.
- Engineered automated emergency vehicle prioritization as measured by path-clearing capabilities for ambulances, by utilizing V2X communication and collaborative platform decision-making.
- Architected Adaptive Traffic Signal Control by processing real-time IoT data directly through the Random Forest algorithm.

PUBLICATIONS & EXTRACURRICULAR

- **Publication:** Authored and published "Multi-Feature Collaborative Traffic Management: An AI And IoT-Integrated Approach", *5th International Conference on Advanced Computational and Communication Paradigms*, Springer, February 2026 (DOI: 10.1007/978-3-032-13544-5_39).
- **Hackathon:** Secured **3rd Prize** in an Intercollegiate Hackathon as measured by competitive ranking at the Smart India Hackathon (SIH) 2025, by developing a Smart Traffic Management System for Urban Congestion.

TECHNICAL SKILLS

- **Languages & Databases:** Python, SQL, HTML, CSS, JavaScript, PHP, Java, C
- **Frameworks & Libraries:** NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn, Streamlit
- **Software & Tools:** PyCharm, Jupyter Notebook, VS Code, Power BI, Tableau, XAMPP, Git